

Lecture 9

Correlation, Regression

Part 2

Prediction, Extrapolation, and Model Complexity

June 2026

INTERPOLATION VS EXTRAPOLATION

Interpolation

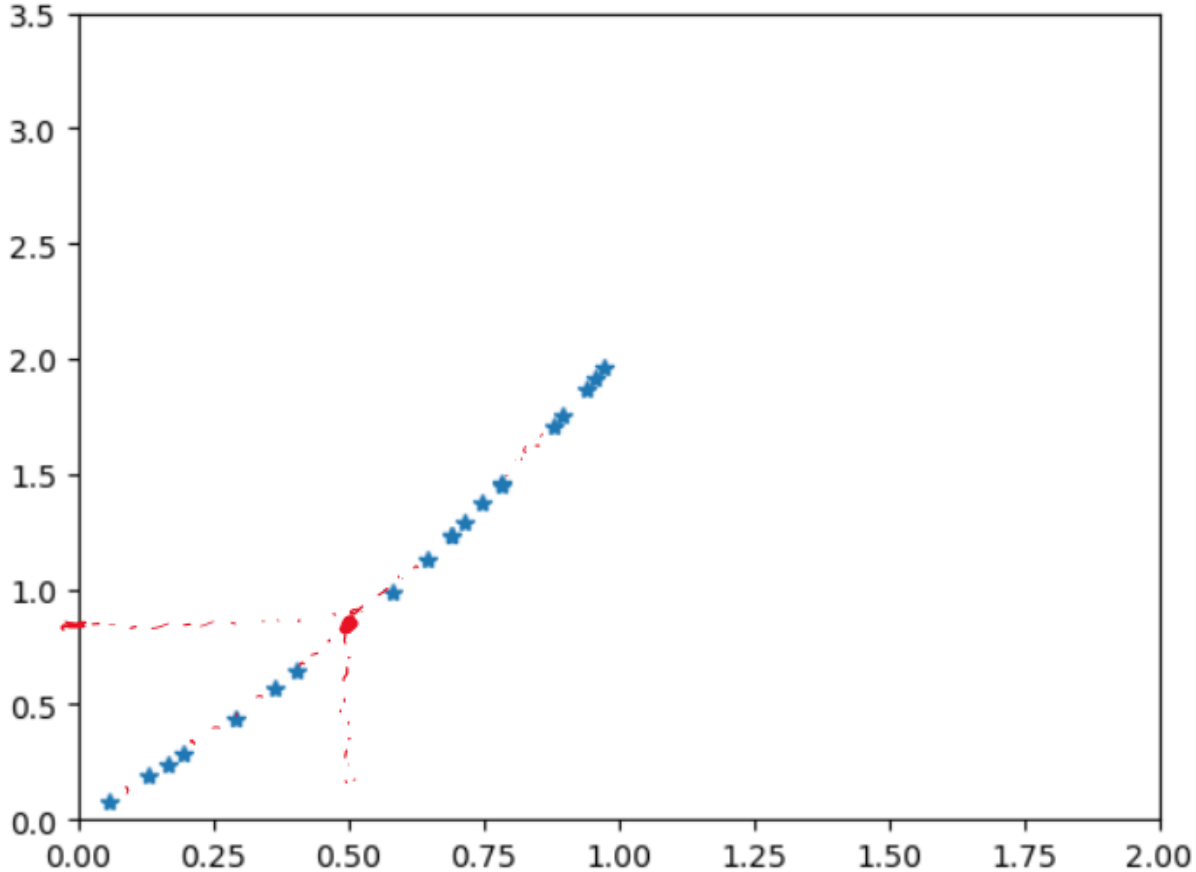
Estimating inside the range of observed data.

Extrapolation

Predicting outside the range of observed data.

The same fitted model can be useful in both cases, but the risk is usually much higher outside the observed range.

INTERPOLATION

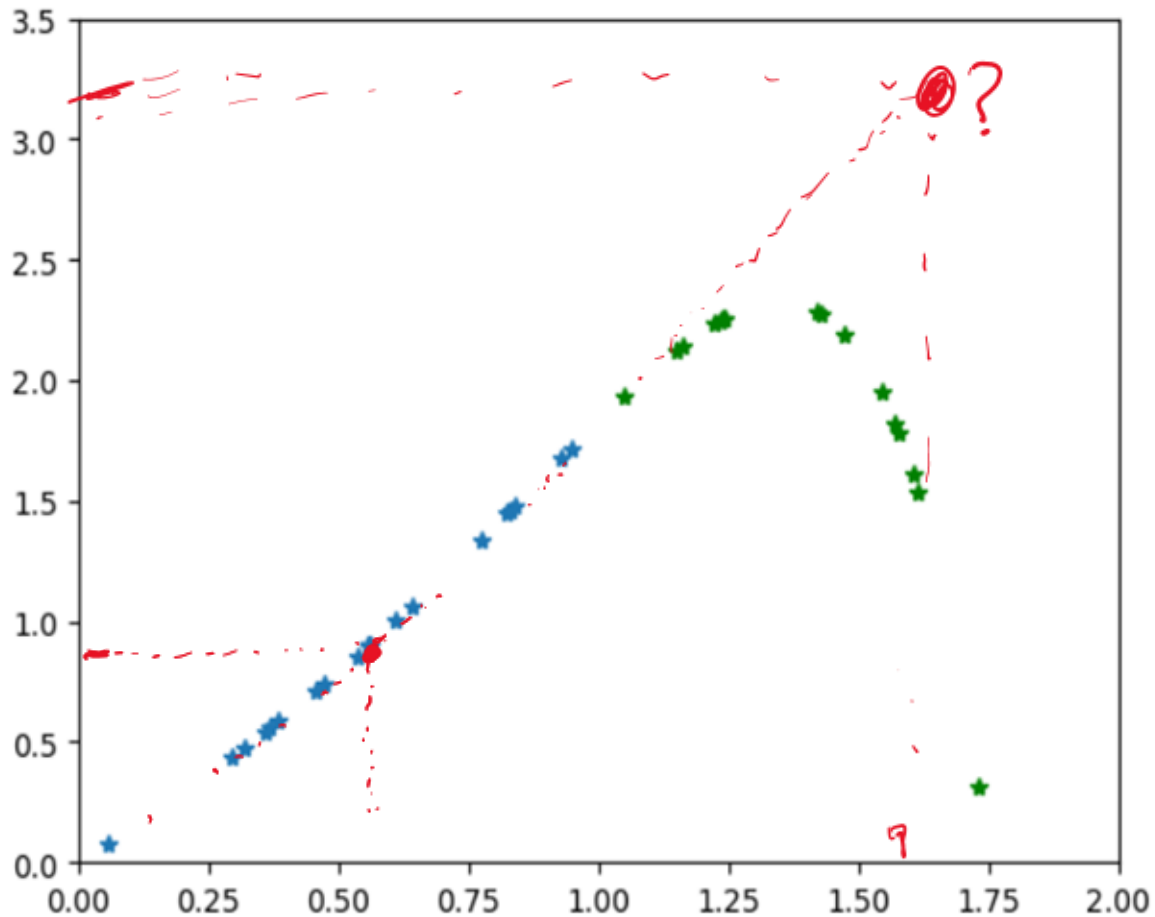


Interpolation estimates values between observed data points.

It is usually safer than extrapolation because nearby observations support the prediction.

It still relies on an assumption: the relationship should not change abruptly between the observed points.

EXTRAPOLATION



Extrapolation predicts beyond the range of observed data.

The model assumes that the same pattern continues outside the data we measured.

This is riskier than interpolation because there are no nearby observations to support the prediction.

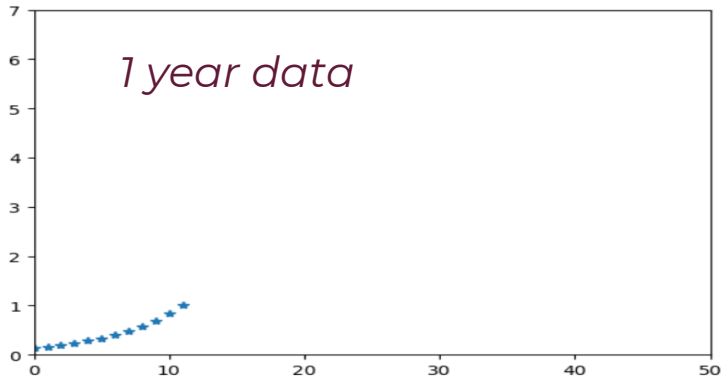
The green points are new measured data.
The extrapolation failed.

THINGS TO CONSIDER

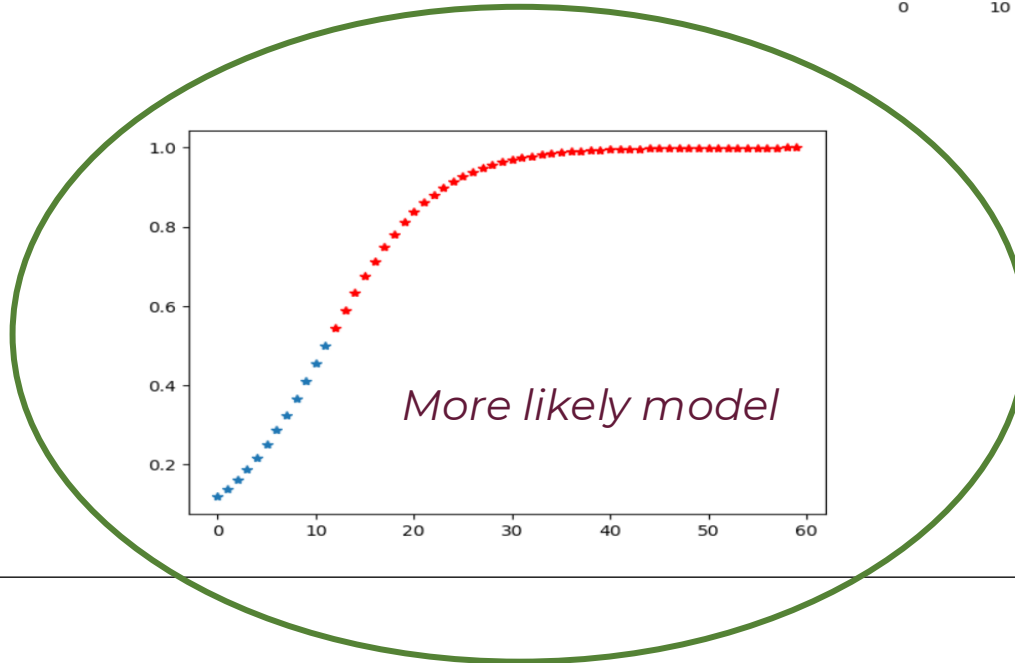
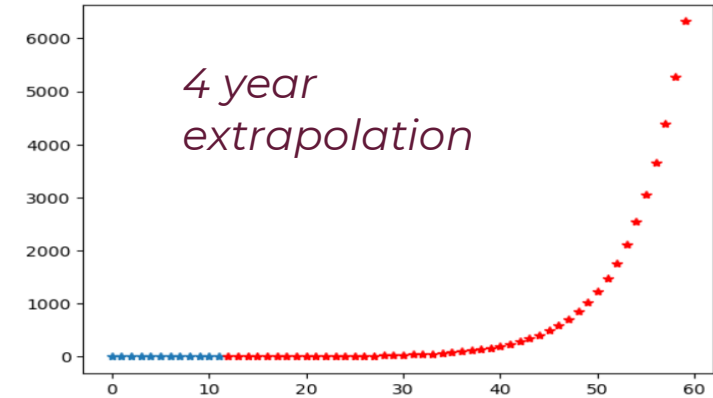
- ❑ Interpolation when we have points nearby is quite safe
- ❑ Extrapolation is risky. We must assume that the model we used for regression was correct and will extend well (and this is not always reasonable)
- ❑ Predicting future data is always extrapolation.
- ❑ When you have many variables, you almost always extrapolate.
- ❑ Growth models never extrapolate indefinitely; they always 'saturate' at some point.

EXAMPLE

A company has 1M followers and has grown by 20% per month. What happens if we extrapolate four years ahead?



Naïve extrapolation will result in an estimate of 6 billion followers !

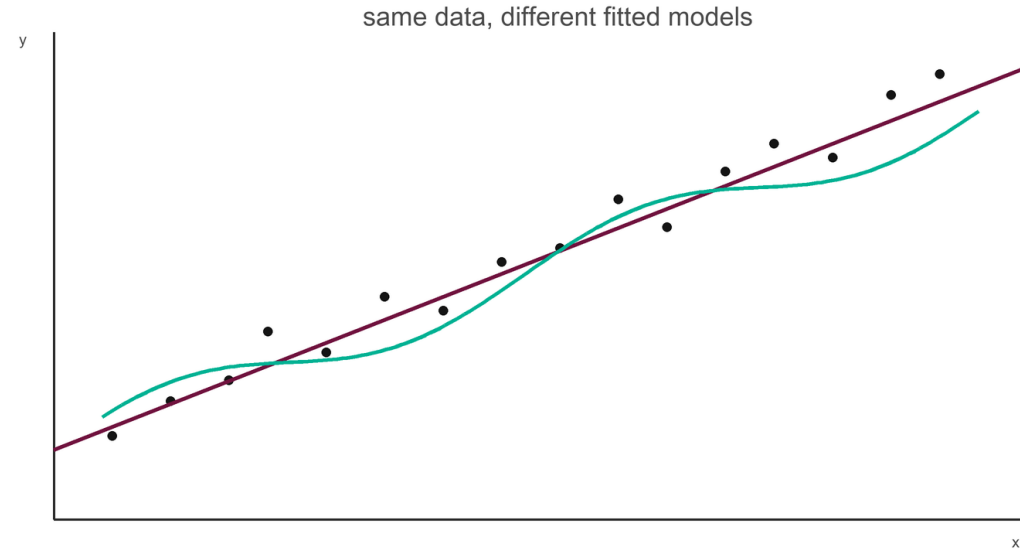


MODEL COMPLEXITY AND PARSIMONY

The same data can often be described by many different models.

A more complex model is more flexible: it can follow more detailed patterns in the sample, but it can also start fitting noise.

Parsimony means choosing the simplest model that captures the main structure of the data well enough for the purpose of the analysis.



LOW COMPLEXITY: HIGH BIAS, LOW VARIANCE

Underfitting

The model is too simple to capture the structure in the data.

High bias

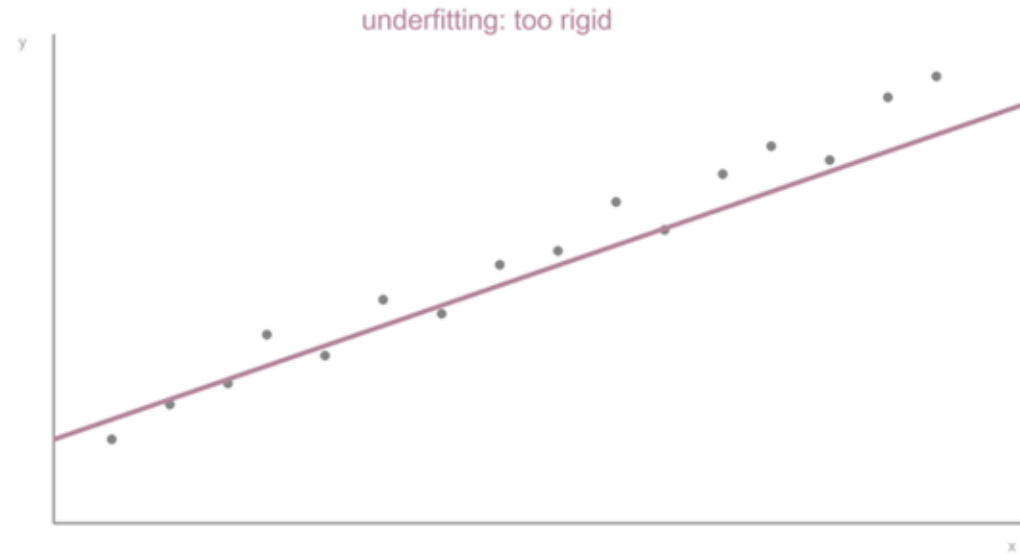
Predictions may be systematically wrong because the model misses part of the relationship.

Low variance

The fitted model is stable: small changes in the data usually lead to similar predictions.

Tradeoff

Low complexity gives stability, but this stability can come at the cost of accuracy.



HIGH COMPLEXITY: LOW BIAS, HIGH VARIANCE

Overfitting

The model follows the training data too closely, including noise.

Low bias

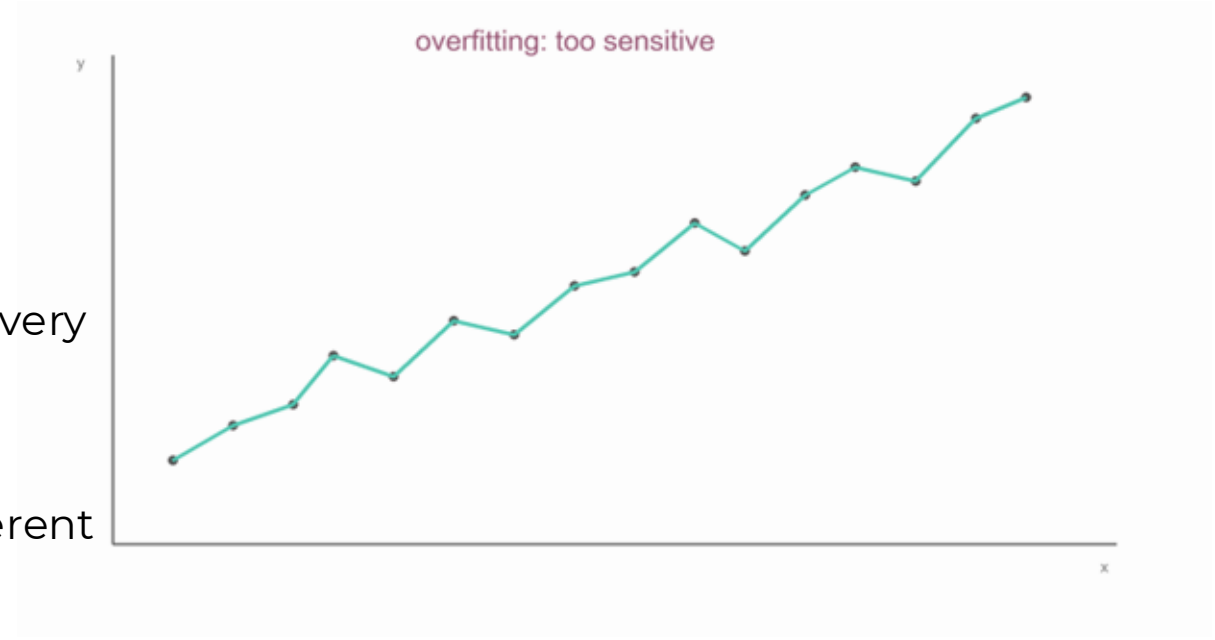
The fitted model can match the training sample very well.

High variance

Small changes in the data can lead to a very different fitted model.

Tradeoff

A very flexible model may look accurate in-sample, but perform poorly on new data.



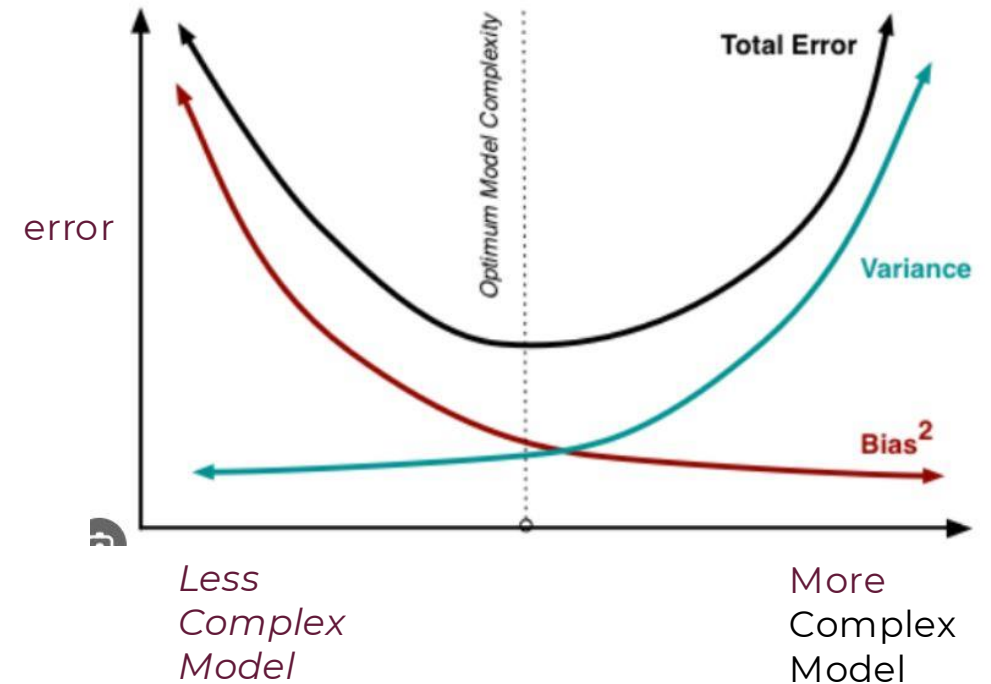
BIAS-VARIANCE TRADE-OFF

More complexity makes a model more flexible.

This can reduce bias because the model can capture structure that a simpler model misses.

But too much flexibility makes the fitted model sensitive to the particular sample. Variance increases because a slightly different sample could produce a different model.

The best model is not the most complex one. It is the model that captures the main structure while still predicting well on new data.



THINGS TO CONSIDER

To test predictive power, we need to evaluate models on data that were not used to fit them. This is called cross-validation.

We also use techniques that reduce model complexity without hurting performance too much. This is called model selection or regularization.

A very complex model that fits the training data well but predicts new scenarios poorly is not very useful.